Inequalities Solutions

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§1 Walkthrough Problems

Problem 1.1. If $abcd = 1$ for a, b, c, $d > 0$, prove that

$$a^4b + b^4c + c^4d + d^4a \geq a + b + c + d.$$

Proof. By Hölder, we have:

$$\left(\sum_{\text{cyc}} a^4b\right)\left(\sum_{\text{cyc}} a\right)\left(\sum_{\text{cyc}} c\right)\left(\sum_{\text{cyc}} d\right) \geq \left(\sum_{\text{cyc}} \sqrt[4]{a^4abcd}\right)^4$$

$$\implies \left(\sum_{\text{cyc}} a^4b\right)(a+b+c+d)^3 \geq (a+b+c+d)^4$$

And we're done. □